

Project Design

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Objective

The objective of this phase is to design the structure of the project and define how users, roles, tables, and ACLs will work together to control record access.

2. System Design

The project is designed around four main components:

- **Users and Roles** – Define who can access and perform operations on records.
- **Tables and Fields** – Store the records and the field values used for access control.
- **Scripted ACLs** – Evaluate access conditions for different record operations.
- **Access Rules** – Determine whether an operation should be allowed or restricted.

3. ACL Design

The project includes four ACL types:

ACL Type	Purpose
-----------------	----------------

Read ACL	Controls viewing of records
----------	-----------------------------

Create ACL	Controls creation of records
------------	------------------------------

Write ACL	Controls modification of records
-----------	----------------------------------

Delete ACL	Controls deletion of records
------------	------------------------------

Each ACL will use the required access conditions and field values to make the access decision.

4. Access Control Flow

User → Role Verification → Record Operation → ACL Triggered → Field Value Checked → Access Granted / Restricted

This flow will be applied to the Read, Create, Write, and Delete operations.

5. Project Design Structure

The overall project design follows this structure:

Users & Roles

↓

Tables & Fields

↓

ACL Configuration

↓

Script-Based Condition Check

↓

Access Decision

↓

Allow / Restrict Operation

6. Conclusion

The Project Design phase defines the overall structure and access-control flow of the system. This design will be used as the foundation for implementing the ACLs and testing their behavior in the following phases.